LOCALLY FINITE ELEMENTARY REPLACEMENTS PRODUCE A DIMENSION-ZERO RESOLUTION OF AN UNBOUNDED COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Before the proof starts, reindex by a finite shift so every term with d(K^n)=infinity lies in degree>=0. Such a shift exists by n+d(K^n)→-infinity as n→-infinity. The initial bounded-below replacement then removes these terms before ξ is asserted finite. Original proof wording is retained.

Editorial scope. One original target with its local construction proved in full. Classical category primitives are explicitly allowed.

Allowed standard prerequisites (editorial).

- Bounded-below resolutions by a monomorphism-cogenerating class. A bounded-below complex embeds quasi-isomorphically in a bounded-below complex whose terms lie in a given class containing zero and into which every object embeds.
- Elementary abelian-category pushouts and local stabilization. A pushout replacement has an acyclic quotient; if each fixed finite-degree window eventually stops changing, cohomology is preserved in the stabilized complex.

Source reference. Original lemma 05T6: focused-seeds/ACL/sources/stacks-snapshot-20261011/derived.tex, source lines 5386–5401. Standard bounded-below replacement used only to initialize; the locally finite unbounded schedule is not given. Stable Stacks tag displayed.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $d : \mathrm{Ob}(\mathcal{A}) \to \{0, 1, 2, \ldots, \infty\}$ be a function. Assume that*

(1) *every object of $\mathcal{A}$ is a subobject of an object A with $d(A) = 0$,*
(2) *$d(A \oplus B) \leq \max\{d(A), d(B)\}$ for $A, B \in \mathcal{A}$, and*
(3) *if $0 \to A \to B \to C \to 0$ is short exact, then $d(C) \leq \max\{d(A) - 1, d(B)\}$.*

Let $K^\bullet$ be a complex such that $n + d(K^n)$ tends to $-\infty$ as $n \to -\infty$. Then there exists a quasi-isomorphism $K^\bullet \to L^\bullet$ with $d(L^n) = 0$ for all $n \in \mathbf{Z}$.

Proof. By Lemma 05T6 we can find a quasi-isomorphism $\sigma_{\geq 0}K^\bullet \to M^\bullet$ with $M^n = 0$ for $n < 0$ and $d(M^n) = 0$ for $n \geq 0$. Then $K^\bullet$ is quasi-isomorphic to the complex

$$\ldots \to K^{-2} \to K^{-1} \to M^0 \to M^1 \to \ldots$$

Hence we may assume that $d(K^n) = 0$ for $n \gg 0$. Note that the condition $n + d(K^n) \to -\infty$ as $n \to -\infty$ is not violated by this replacement.

We are going to improve $K^\bullet$ by an (infinite) sequence of elementary replacements. An *elementary replacement* is the following. Choose an index n such that $d(K^n) > 0$. Choose an injection $K^n \to M$ where $d(M) = 0$. Set $M' = \mathrm{Coker}(K^n \to$

Date: stacks-project:derived.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

$M \oplus K^{n+1}$). Consider the map of complexes

$$\begin{array}{ccccccccc} K^\bullet : & & K^{n-1} & \longrightarrow & K^n & \longrightarrow & K^{n+1} & \longrightarrow & K^{n+2} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ (K')^\bullet : & & K^{n-1} & \longrightarrow & M & \longrightarrow & M' & \longrightarrow & K^{n+2} \end{array}$$

It is clear that $K^\bullet \to (K')^\bullet$ is a quasi-isomorphism. Moreover, it is clear that $d((K')^n) = 0$ and

$$d((K')^{n+1}) \leq \max\{d(K^n) - 1, d(M \oplus K^{n+1})\} \leq \max\{d(K^n) - 1, d(K^{n+1})\}$$

and the other values are unchanged.

To finish the proof we carefully choose the order in which to do the elementary replacements so that for every integer m the complex $\sigma_{\geq m} K^\bullet$ is changed only a finite number of times. To do this set

$$\xi(K^\bullet) = \max\{n + d(K^n) \mid d(K^n) > 0\}$$

and

$$I = \{n \in \mathbf{Z} \mid \xi(K^\bullet) = n + d(K^n) \text{ and } d(K^n) > 0\}$$

Our assumption that $n + d(K^n)$ tends to $-\infty$ as $n \to -\infty$ and the fact that $d(K^n) = 0$ for $n >> 0$ implies $\xi(K^\bullet) < +\infty$ and that I is a finite set. It is clear that $\xi((K')^\bullet) \leq \xi(K^\bullet)$ for an elementary transformation as above. An elementary transformation changes the complex in degrees $\leq \xi(K^\bullet) + 1$. Hence if we can find finite sequence of elementary transformations which decrease $\xi(K^\bullet)$, then we win. However, note that if we do an elementary transformation starting with the smallest element $n \in I$, then we either decrease the size of I, or we increase $\min I$. Since every element of I is $\leq \xi(K^\bullet)$ we see that we win after a finite number of steps. $\square$